Student Performance Prediction System

Phase 2 Report: Data Preprocessing, Feature Selection & Dimensionality Reduction

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Member	Role	University Roll No.
Member 1	Team Leader (Arjun Bhandana)	_____2028660_____
Member 2	Team Member(Sarthak Gera)	_____2028464_____
Member 3	Arsh Singh	————2028663—————

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1. Problem Description

The Student Performance Prediction System is a Python-based machine learning application developed to automate the analysis and prediction of student academic outcomes. It streamlines the process of collecting student data, training predictive models, and generating performance insights while maintaining accuracy and efficiency.

Phase 2 focuses on the critical preprocessing pipeline applied to the cleaned dataset from Phase 1. This includes feature scaling, target variable engineering, feature selection using sequential methods, and dimensionality reduction using Principal Component Analysis (PCA). The outputs of this phase — a processed dataset and a PCA-transformed dataset — form the foundation for model training in Phase 3.

2. Architecture / Pipeline Overview

The Phase 2 pipeline consists of the following sequential stages:

Step	Stage	Description
1	**Load Cleaned Dataset**	Read Cleaned_Dataset.csv containing 9 features including Study_Hours, Attendance_Percentage, Internal_Assessment_Score, Previous_Academic_Result, Final_Grade, Study_Efficiency, Internet_Usage_Hours, Gender_Male, and Result.
2	**Feature Scaling**	Apply both MinMaxScaler (normalizes to [0,1]) and StandardScaler (zero mean, unit variance) to the feature matrix X (all columns except Final_Grade).
3	**Target Engineering**	Create binary classification label 'Result' (1 = Pass if Final_Grade ≥ 50, else 0) and a multi-class 'Grade_Label' (A/B/C/D/F).
4	**Feature Selection**	Apply Forward and Backward Sequential Feature Selection (SFS) using Linear Regression as estimator to select the top 3 most predictive features.

5	**Dimensionality Reduction**	Apply PCA on the standardized feature matrix to reduce dimensions while retaining ≥ 95% of explained variance.
6	**Export Datasets**	Save processed_data.csv (full engineered dataset) and pca_data.csv (PCA-transformed features) for use in Phase 3.

3. Preprocessing Methods Applied

3.1 Feature Scaling

Two scaling strategies were applied to the feature matrix (all columns except Final_Grade, Result, Result_Label, Grade_Label):

- MinMaxScaler: Scales each feature to the range [0, 1]. Formula: x' = (x - x_min) / (x_max - x_min). Useful for algorithms sensitive to the magnitude of values.
- StandardScaler: Transforms features to have zero mean and unit variance. Formula: x' = (x - mean) / std. Used as input for PCA and SFS, which are variance-sensitive.

3.2 Target Variable Engineering

Two new target columns were derived from Final_Grade:

- Result (Binary): 1 (Pass) if Final_Grade >= 50, else 0 (Fail). Supports binary classification tasks.
- Grade_Label (Multi-class): A (>= 85), B (>= 75), C (>= 60), D (>= 40), F (< 40). Supports grade category classification.

3.3 Feature Selection — Sequential Feature Selector (SFS)

scikit-learn's SequentialFeatureSelector was used with LinearRegression as the underlying estimator to select the 3 most predictive features for Final_Grade:

- Forward Selection: Starts with no features, adds the best one at each step. Result: ['Study_Hours', 'Internal_Assessment_Score', 'Previous_Academic_Result']
- Backward Elimination: Starts with all features, removes the weakest one at each step. Result: ['Study_Hours', 'Internal_Assessment_Score', 'Previous_Academic_Result']

Both methods converged on the same 3 features, confirming their importance for predicting student performance.

3.4 Dimensionality Reduction — Principal Component Analysis (PCA)

PCA was applied on the StandardScaler-normalized feature matrix. Key results:

Component	Explained Variance	Cumulative Variance	Retained?
PC1	24.20%	24.20%	Yes
PC2	14.78%	38.98%	Yes
PC3	14.50%	53.48%	Yes
PC4	14.30%	67.78%	Yes
PC5	14.23%	82.01%	Yes
PC6	13.76%	95.76%	Yes (threshold met)
PC7	4.24%	100.00%	No (excluded)

6 principal components were retained (capturing 95.76% of total variance), reducing dimensionality from 7 input features while preserving almost all information content. The PCA-projected dataset was saved as pca_data.csv.

4. Evaluation Metrics Used

Phase 2 is a preprocessing and feature engineering phase, so standard prediction metrics (accuracy, F1, etc.) are not directly applicable. Instead, the following quality indicators were used:

Metric / Criterion	Applied To	Result
Feature Agreement (Forward vs Backward SFS)	Feature Selection	Both methods selected identical features — strong validation
Explained Variance Ratio	PCA	≥ 95% variance retained with 6 components
Cumulative Variance Threshold	PCA	Used to determine optimal number of components

Data Consistency Check	Label Engineering	Binary Result and Grade_Label consistent with Final_Grade values
Scaler Output Verification	MinMaxScaler / StandardScaler	Sample output rows verified for correct range/distribution

5. Preliminary Results and Visualizations

5.1 Dataset Overview (after Engineering)

After Phase 2 processing, the dataset contains the following columns:

Column	Type	Description
Study_Hours	Float	Daily study hours (input feature)
Attendance_Percentage	Float	Attendance rate (input feature)
Internal_Assessment_Score	Float	IA marks (input feature)
Previous_Academic_Result	Float	Prior academic score (input feature)
Final_Grade	Integer	Target for regression (original)
Study_Efficiency	Float	Engineered ratio (input feature)
Internet_Usage_Hours	Float	Internet usage (input feature)
Gender_Male	Binary	Gender encoding (0/1)
Result	Binary	1=Pass (Final_Grade>=50), 0=Fail
Result_Label	String	Human-readable Pass/Fail label
Grade_Label	String	Multi-class grade: A/B/C/D/F

5.2 Sample Data (First 5 Records — Original)

Sample records from the cleaned input dataset before engineering:

Study_Hrs	Attendance%	IA_Score	Prev_Result	Final_Grade	Efficiency	Internet_Hrs
6.7	74.9	41.4	83.8	75	0.67	6.5
9.6	73.3	41.9	76.4	40	0.74	5.8
7.6	67.0	40.1	60.8	60	0.77	4.1
6.4	84.3	49.5	79.8	75	0.85	5.6
2.4	80.7	33.4	76.4	60	0.93	2.7

5.3 Feature Selection Results

Forward Selection (n=3 features): ['Study_Hours', 'Internal_Assessment_Score', 'Previous_Academic_Result']

Backward Elimination (n=3 features): ['Study_Hours', 'Internal_Assessment_Score', 'Previous_Academic_Result']

Both methods independently selected the same three features, confirming that Study_Hours, Internal_Assessment_Score, and Previous_Academic_Result are the strongest predictors of Final_Grade in this dataset.

5.4 PCA Visualization — PC1 vs PC2 Scatter Plot

A 2D scatter plot of the first two principal components (PC1 vs PC2) was generated using matplotlib to visualize the distribution of data points in the reduced feature space. The plot reveals the spread and structure of the student dataset after dimensionality reduction, and can be used to visually detect clusters or patterns. [See Fig. 2 — PCA Projection scatter plot generated in the Jupyter Notebook].

5.5 Output Datasets

- processed_data.csv — Full dataset with original features plus engineered columns (Result, Result_Label, Grade_Label).
- pca_data.csv — PCA-transformed dataset with 6 principal components (PC1–PC6) retaining 95.76% of variance.

6. References

Books:

- Let Us Python – Yashavant Kanetkar

Libraries & Documentation:

- scikit-learn Documentation — SequentialFeatureSelector, PCA, MinMaxScaler, StandardScaler: https://scikit-learn.org/
- pandas Documentation: https://pandas.pydata.org/
- matplotlib Documentation: https://matplotlib.org/

Online Resources:

- GeeksforGeeks – Feature Selection, PCA: https://www.geeksforgeeks.org/
- TutorialsPoint – Machine Learning: https://www.tutorialspoint.com/
- Stack Overflow – https://stackoverflow.com